Applications:
An Overview

FLORIDA MUSEUM

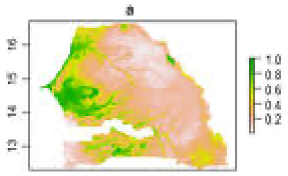
UF UNIVERSITY of FLORIDA

iDigBio
Integrated Digitized Biocollections

NSF

Connecting Specimens, Trees, Niche Models, Tools: Examples

RESOURCES:

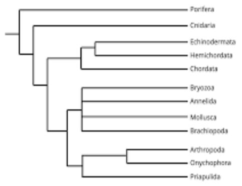

a
16
15
14
13
1.0
0.8
0.6
0.4
0.2

N

Niche Models
- ecological niche modeling
- biodiversity and range analysis
- visualization

A

Analytical Tools
- evolutionary models
- comparative methods
- visualization


Porifera
Cnidaria
Echinodermata
Hemichordata
Chordata
Bryozoa
Annelida
Mollusca
Brachiopoda
Arthropoda
Onychophora
Priapulida

T

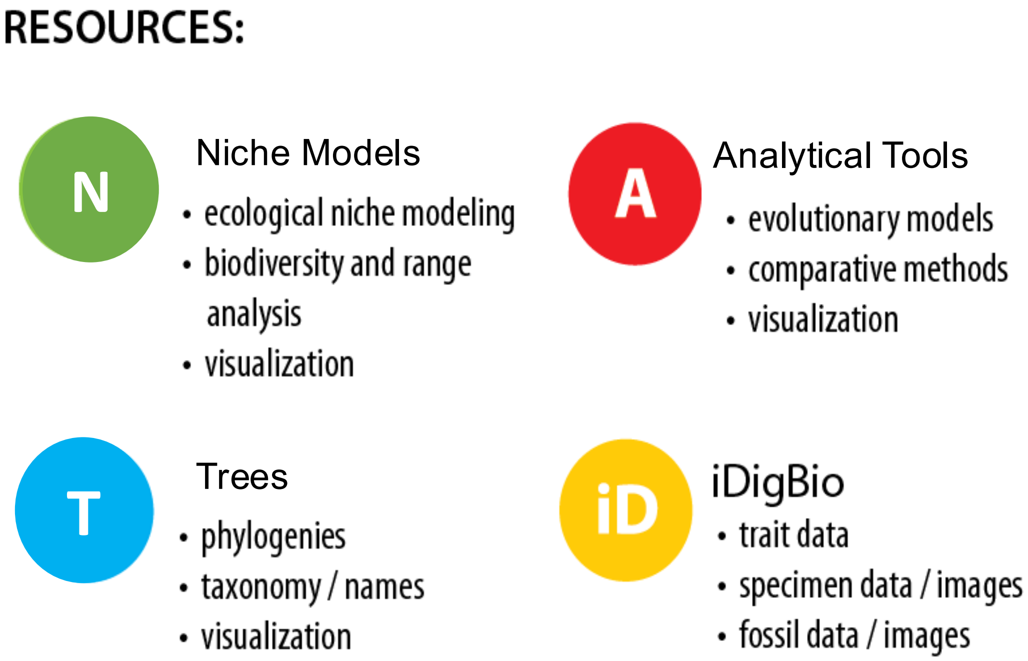
Trees
- phylogenies
- taxonomy / names
- visualization

iD

iDigBio
- trait data
- specimen data / images
- fossil data / images

iDigBio
Integrated Digitized Biocollections

Connecting Specimens, Trees, Niche Models, Tools: Examples



RESOURCES:

N Niche Models
- ecological niche modeling
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- visualization

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T Trees
- phylogenies
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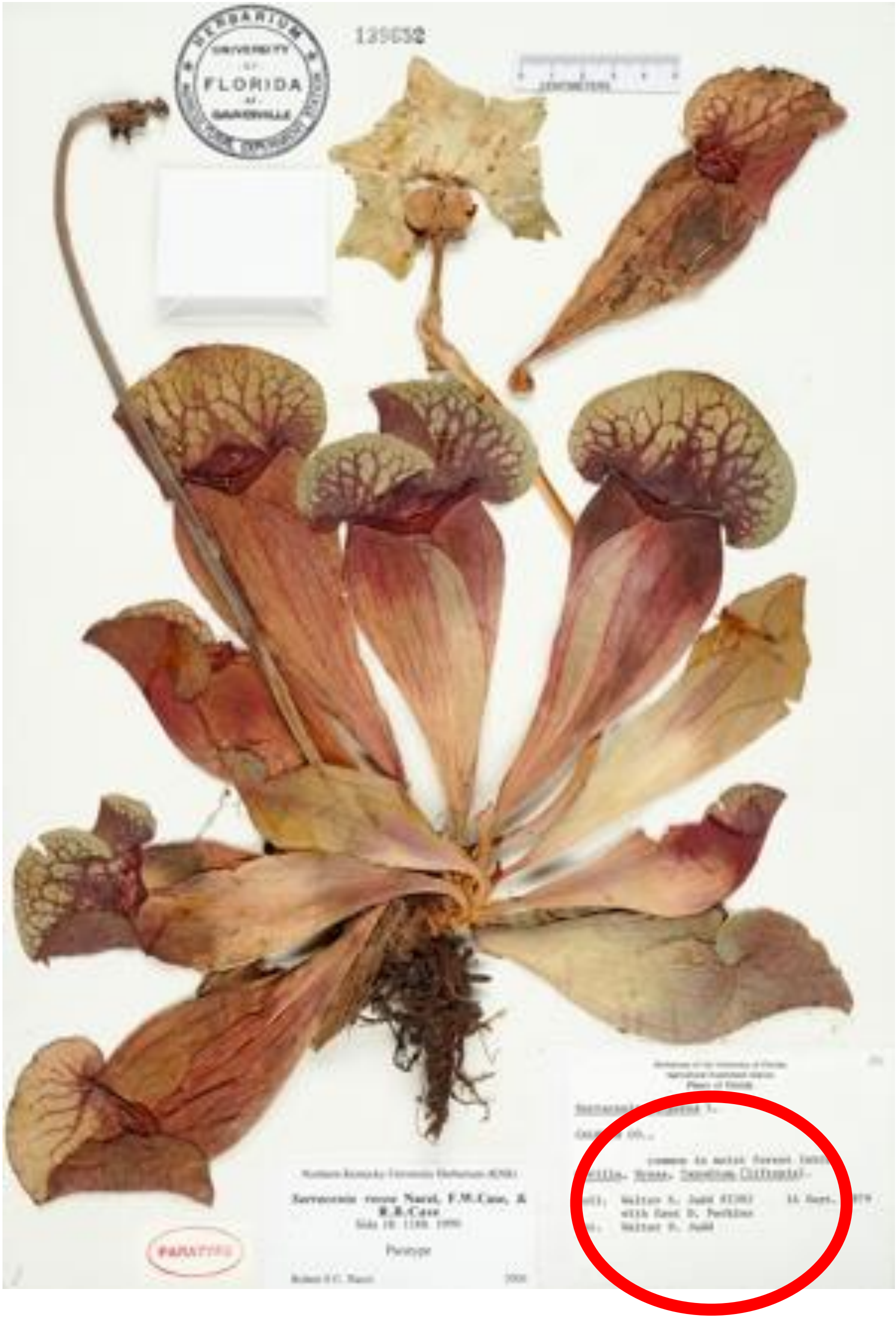
iD iDigBio
- trait data
- specimen data / images
- fossil data / images

Allen et al.. 2019. Biodiversity synthesis across the green branches of the tree of life. *Nature Plants* 5:11-13.

Label Data (Occurrences)

- Scientific name
- Date
- Collector
- Location – state, county,
  specific site,
  GPS coordinates
- Associated species

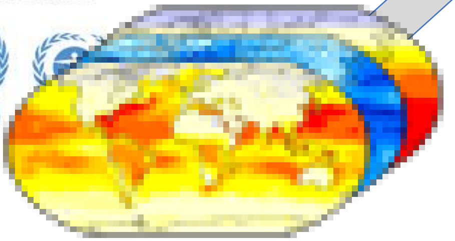


Niche Models

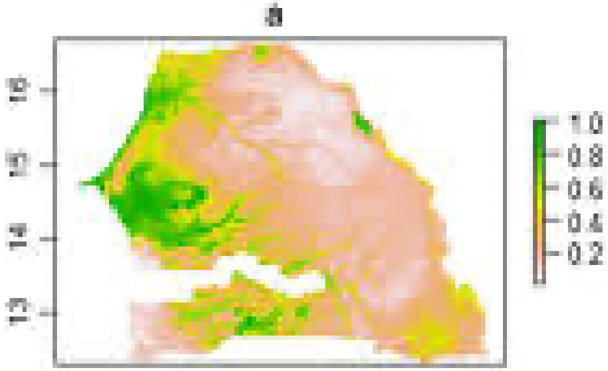
Species Occurrence Data

IPCC
INTERGOVERNMENTAL
PANEL ON
CLIMATE CHANGE

WMO



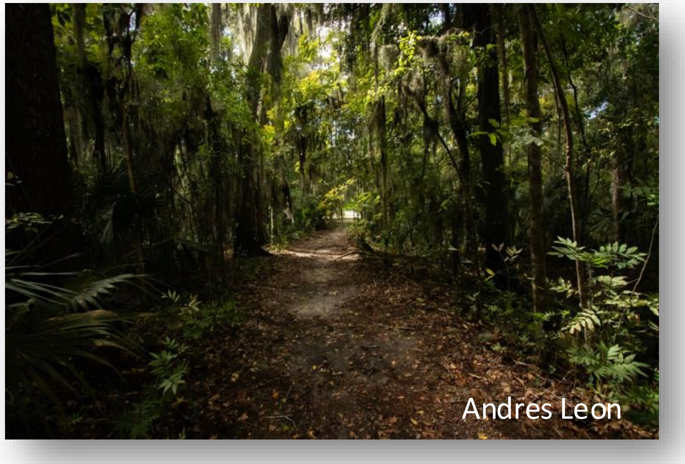
Environmental Data


a
16
15
14
13
1.0
0.8
0.6
0.4
0.2

Predicted Habitat

FLORIDA UPLAND HARDWOOD FORESTS

John Rouse



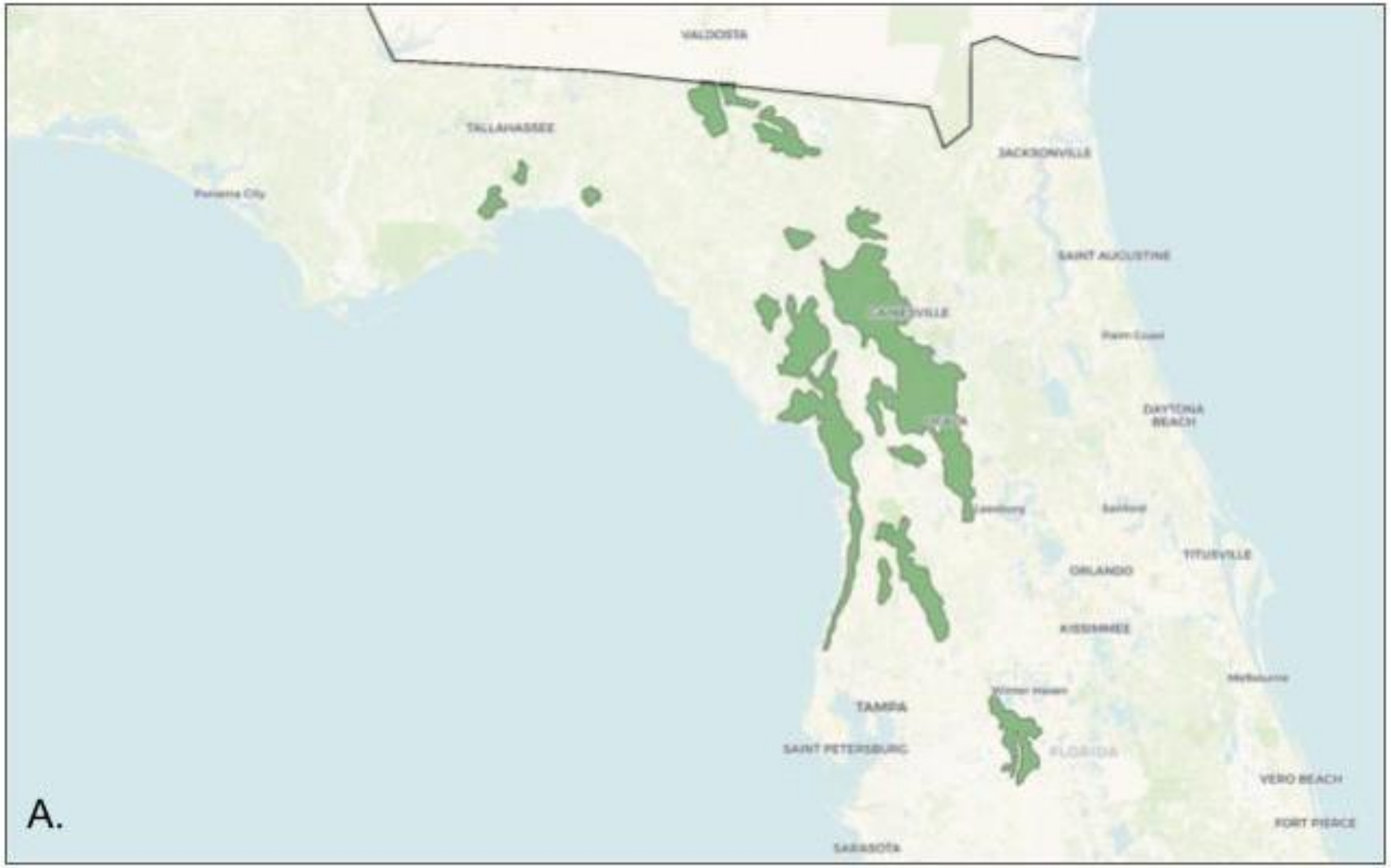
Andres Leon

Mariela Pajuelo

McCarty Woods, University of Florida

FLORIDA UPLAND HARDWOOD FORESTS

Florida
Natural Areas
Inventory


VALDOSTA
TALLAHASSEE
Panama City
JACKSONVILLE
SAINT AUGUSTINE
GAINESVILLE
Palm Coast
DAYTONA BEACH
OCALA
Leesburg
Sanford
TITUSVILLE
ORLANDO
KISSIMMEE
Winter Haven
Melbourne
TAMPA
SAINT PETERSBURG
FLORIDA
VERO BEACH
FORT PIERCE
SARASOTA
A.

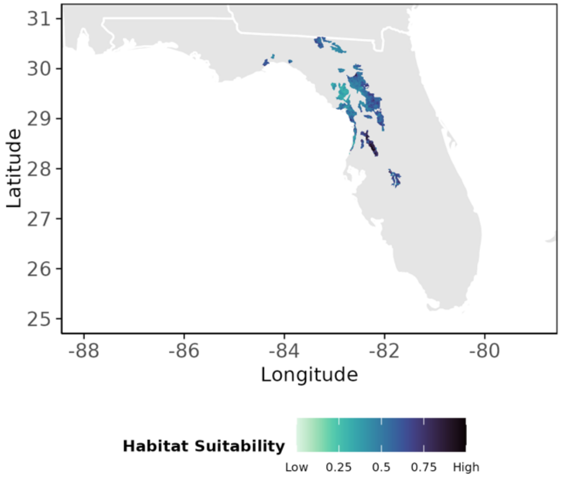
- 1,248,796 occurrence records for 39 focal species and 32 plant families.
- Final cleaned dataset contained 371,253 occurrence records.

FLORIDA UPLAND HARDWOOD FORESTS | *Future Habitat Suitability*

- *Tilia americana* and many focal taxa reach southern edge of distribution in Florida.
- Projected to lose habitat suitability in Florida Upland Hardwoods
- Future projections also indicate broad retention of suitable habitat across its core range.
- Results suggest relative climatic stability outside of Florida.

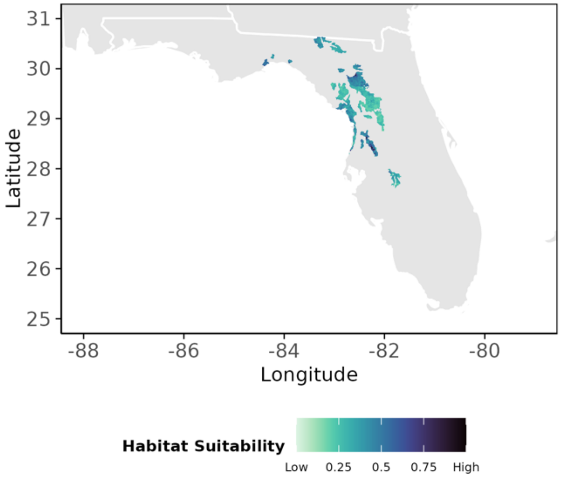
Tilia americana Florida Hardwood Forests

Current Habitat Suitability (70.69%)


Latitude
31
30
29
28
27
26
25
-88
-86
-84
-82
-80
Longitude
Habitat Suitability
Low
0.25
0.5
0.75
High

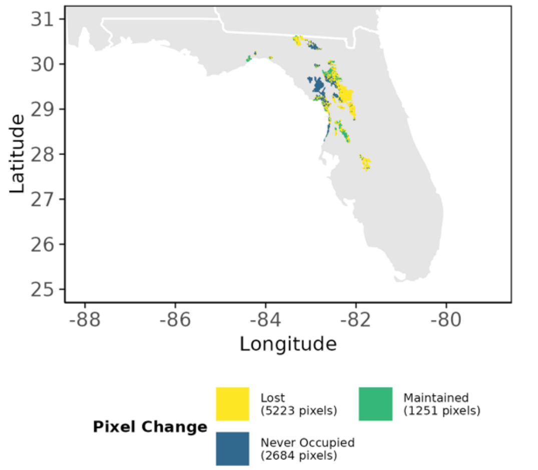
Tilia americana Florida Hardwood Forests

Future Habitat Suitability (13.66%)


Latitude
31
30
29
28
27
26
25
-88
-86
-84
-82
-80
Longitude
Habitat Suitability
Low
0.25
0.5
0.75
High

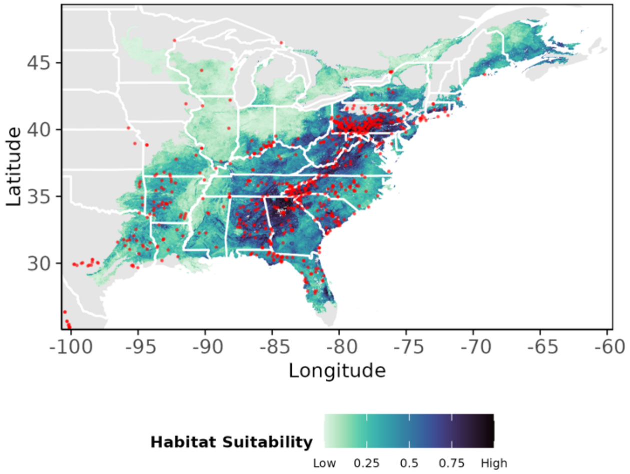
Tilia americana Florida Hardwood Forests

Change in Habitat Suitability (1 pixel ~ 1 km^2)

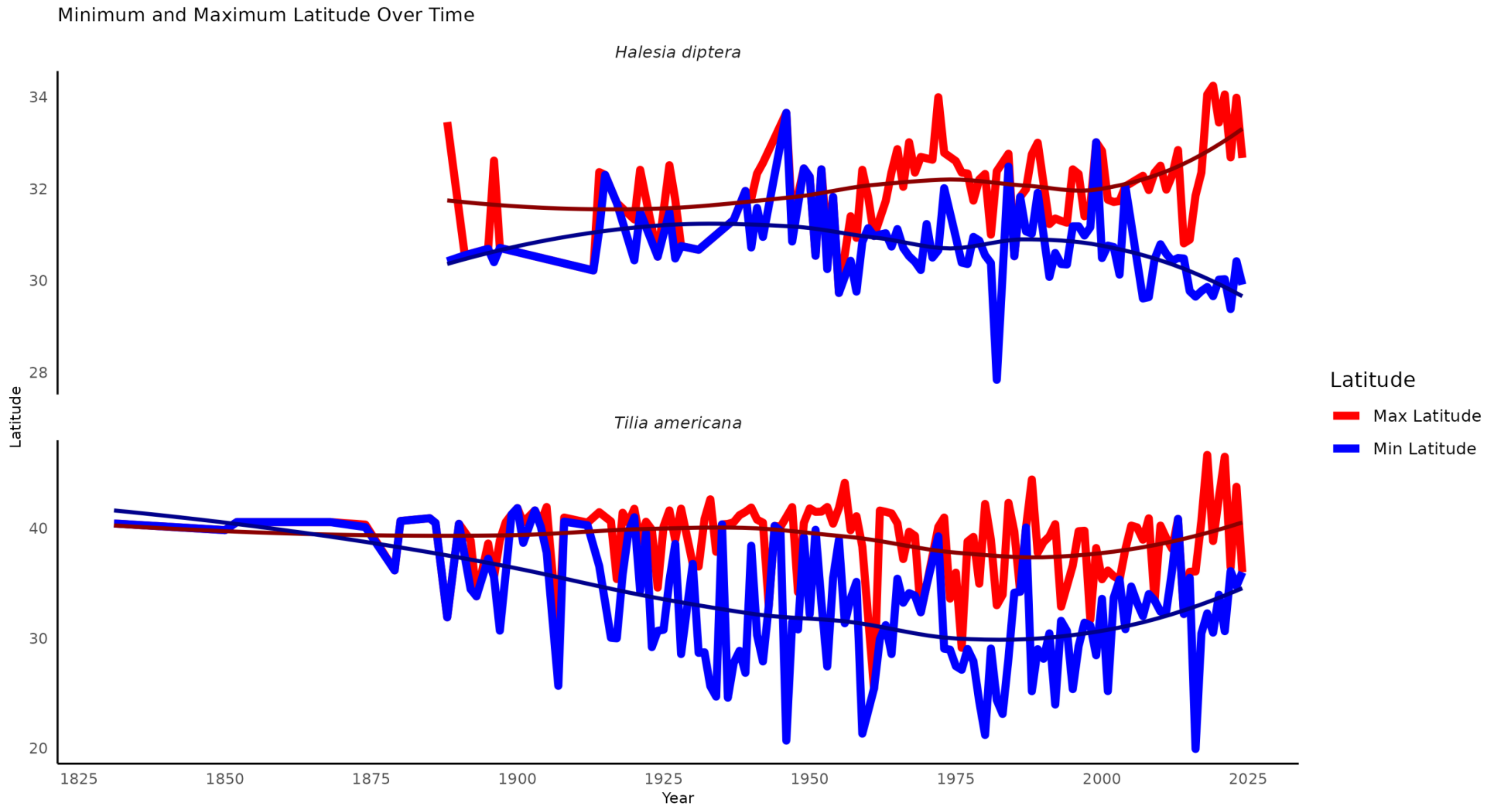

Latitude
31
30
29
28
27
26
25
-88
-86
-84
-82
-80
Longitude
Pixel Change
Lost (5223 pixels)
Maintained (1251 pixels)
Never Occupied (2684 pixels)

Tilia americana Eastern Temperate Forests

Future Habitat Suitability

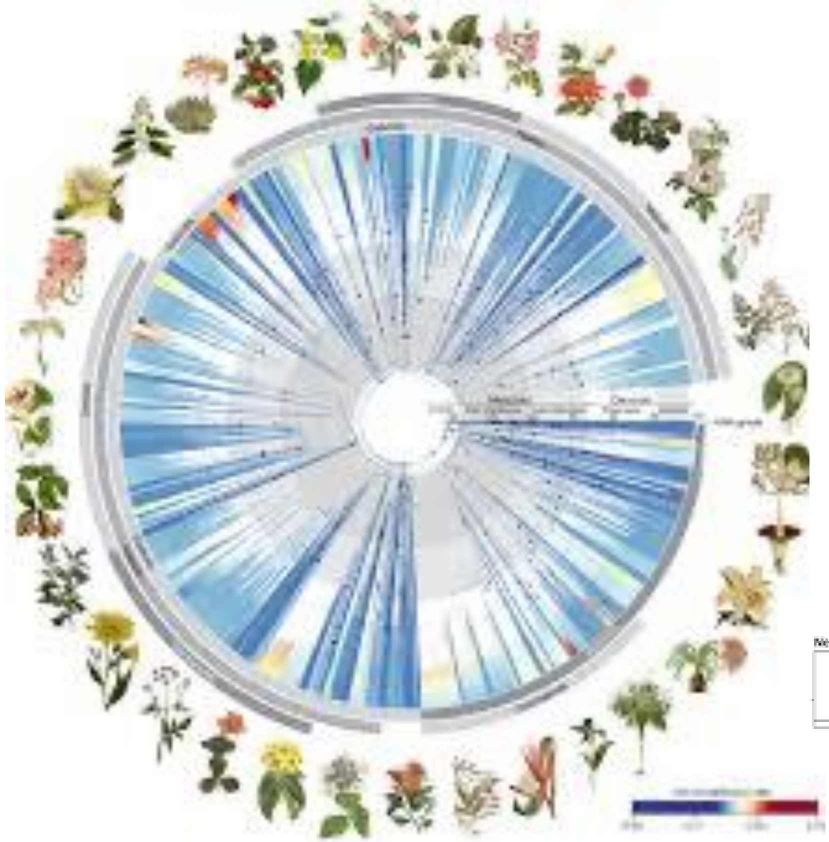

Latitude
45
40
35
30
-100
-95
-90
-85
-80
-75
-70
-65
-60
Longitude
Habitat Suitability
Low
0.25
0.5
0.75
High

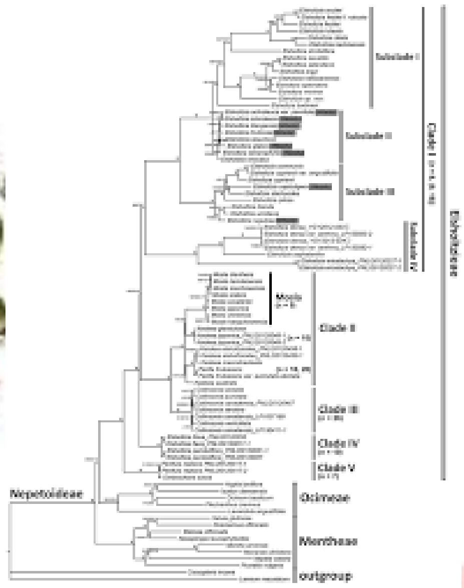
FLORIDA UPLAND HARDWOOD FORESTS | *Latitudinal Range Change*

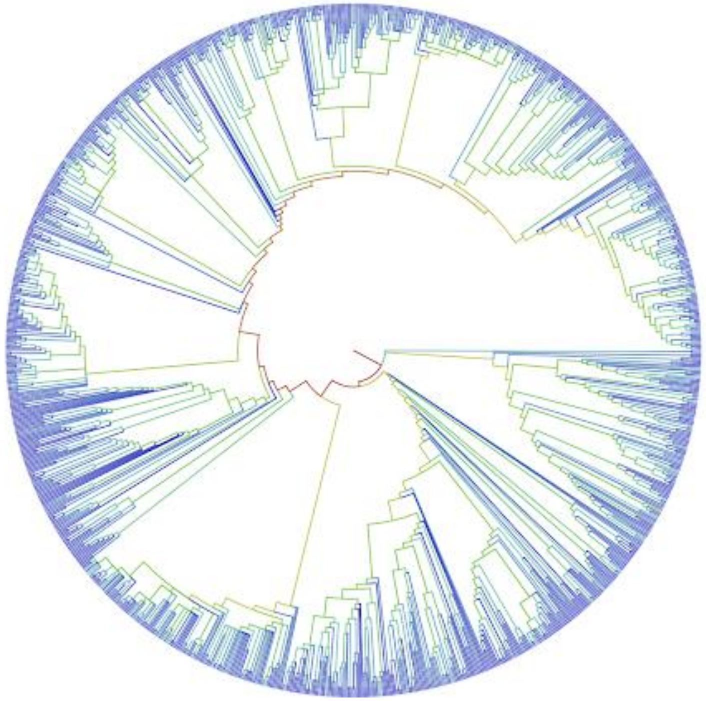


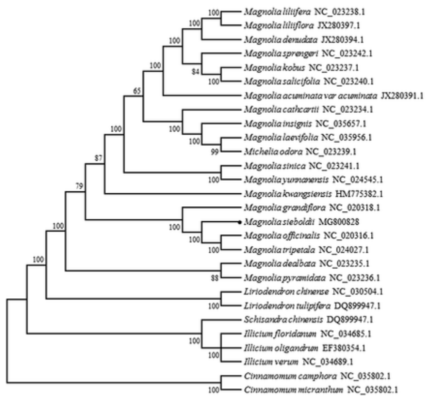
- *Halesia diptera* showed a widening latitudinal range over time.
- *Tilia americana* showed a northward retreat of its southern range limit, suggesting trailing-edge contraction.

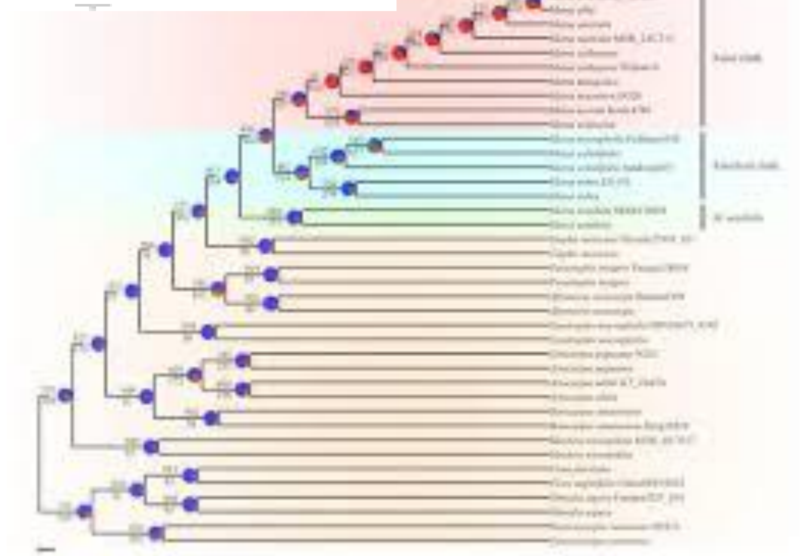
Trees




Subclade I
Subclade II
Subclade III
Subclade IV
Clade I
Elsholtzieae
Mosla
Clade II
Clade III
Clade IV
Clade V
Nepetoideae
Ocimeae
Mentheae
outgroup




Magnolia lilitifera NC_023238.1
Magnolia lilitiflora JX280397.1
Magnolia denudata JX280394.1
Magnolia sprengeri NC_023242.1
Magnolia kobus NC_023237.1
Magnolia salicifolia NC_023240.1
Magnolia acuminata var acuminata JX280391.1
Magnolia cathcartii NC_023234.1
Magnolia insignis NC_035657.1
Magnolia laevifolia NC_035956.1
Michelia odora NC_023239.1
Magnolia sinica NC_023241.1
Magnolia yunnanensis NC_024545.1
Magnolia kwangsiensis HM775382.1
Magnolia grandiflora NC_020318.1
Magnolia sieboldii MG800828
Magnolia officinalis NC_020316.1
Magnolia tripetala NC_024027.1
Magnolia dealbata NC_023235.1
Magnolia pyramidata NC_023236.1
Liriodendron chinense NC_030504.1
Liriodendron tulipifera DQ899947.1
Schisandra chinensis DQ899947.1
Illicium floridanum NC_034685.1
Illicium oligandrum EF380354.1
Illicium verum NC_034689.1
Cinnamomum camphora NC_035802.1
Cinnamomum micranthum NC_035802.1



Ancestral Niche

Connecting Trees, Specimens, Tools

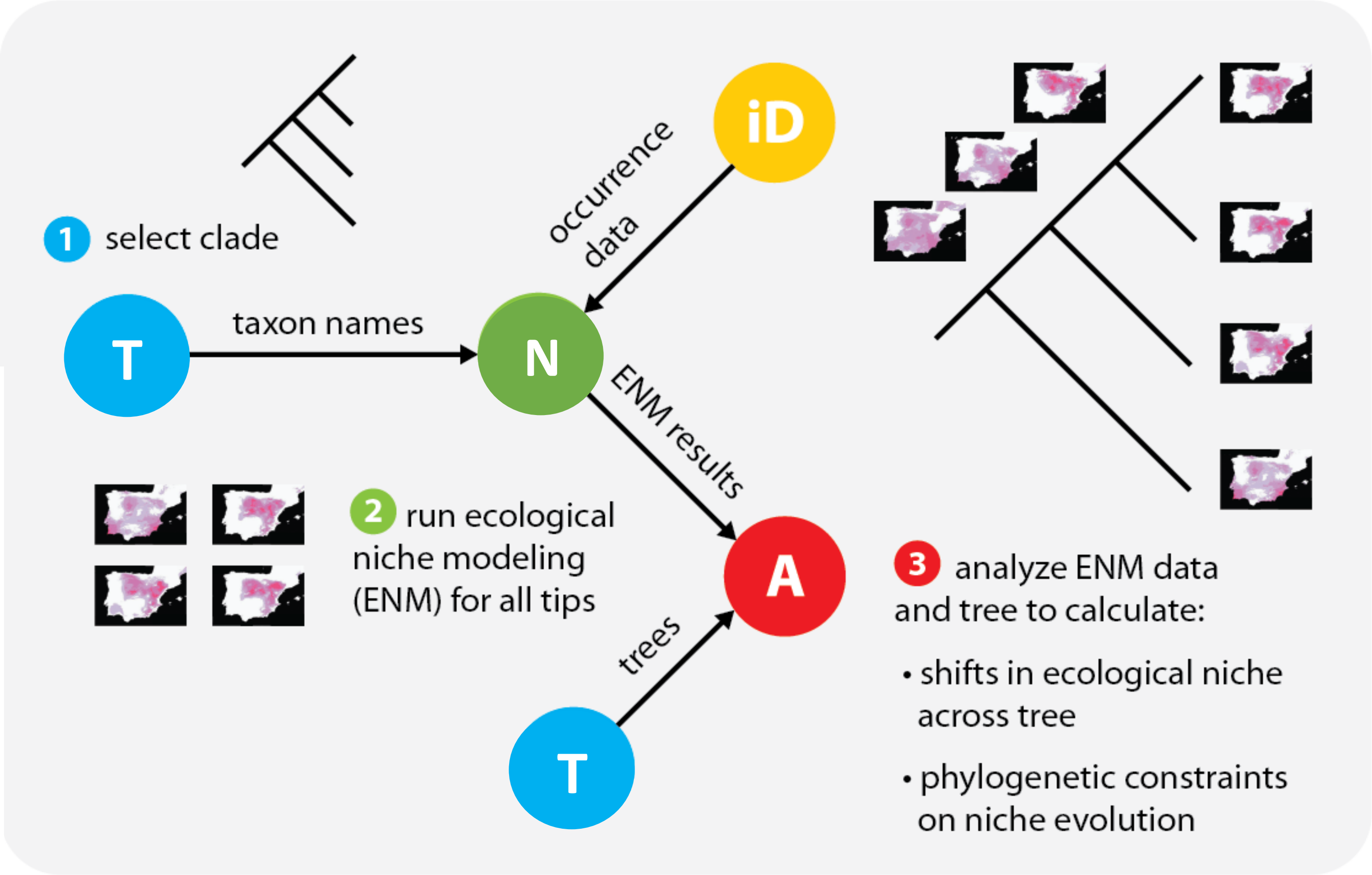
RESOURCES:

N Niche Models
- ecological niche modeling
- biodiversity and range analysis
- visualization

A Analytical Tools
- evolutionary models
- comparative methods
- visualization

T Trees
- phylogenies
- taxonomy / names
- visualization

iD iDigBio
- trait data
- specimen data / images
- fossil data / images

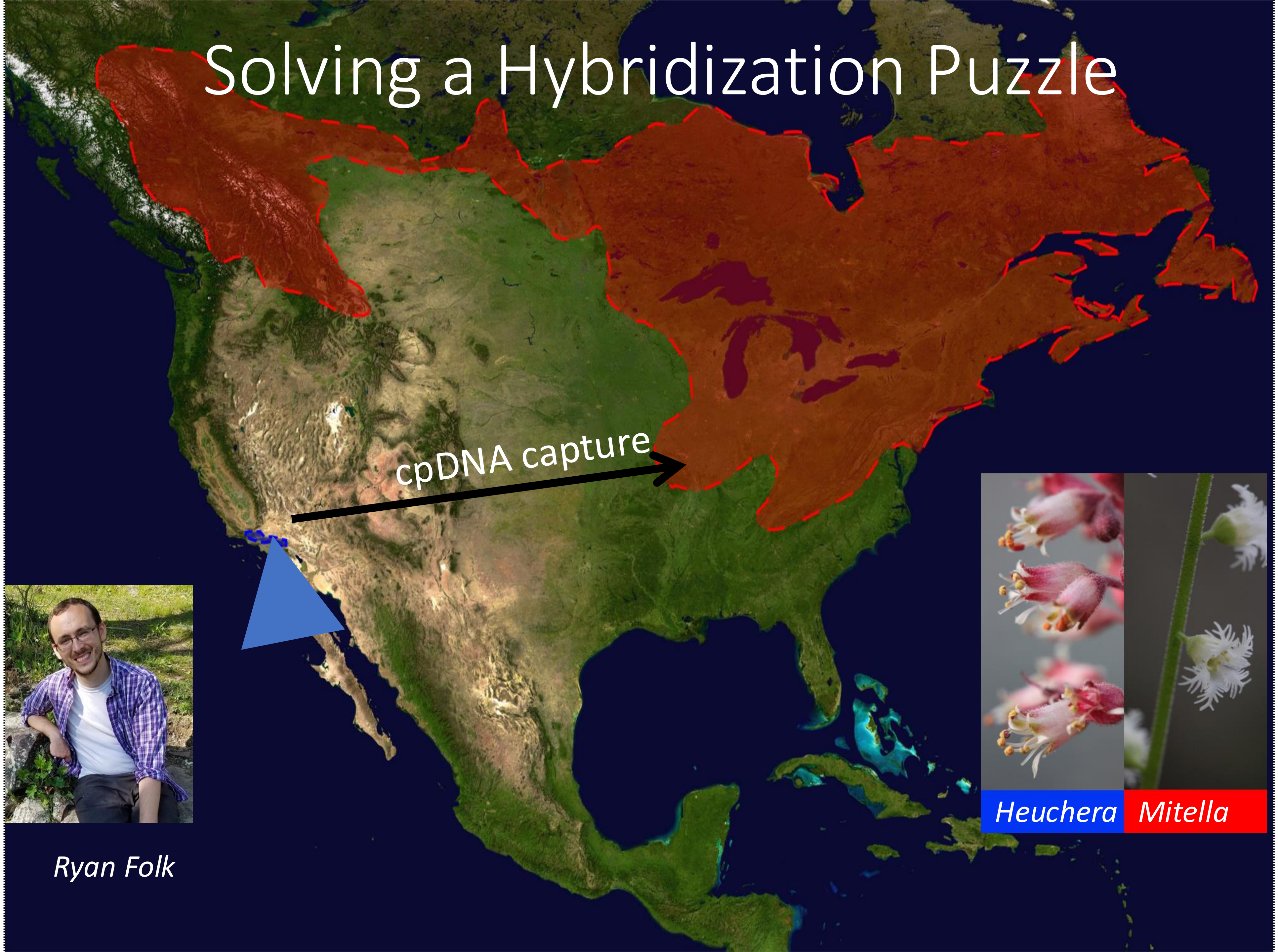


① select clade

② run ecological niche modeling (ENM) for all tips

③ analyze ENM data and tree to calculate:
- shifts in ecological niche across tree
- phylogenetic constraints on niche evolution

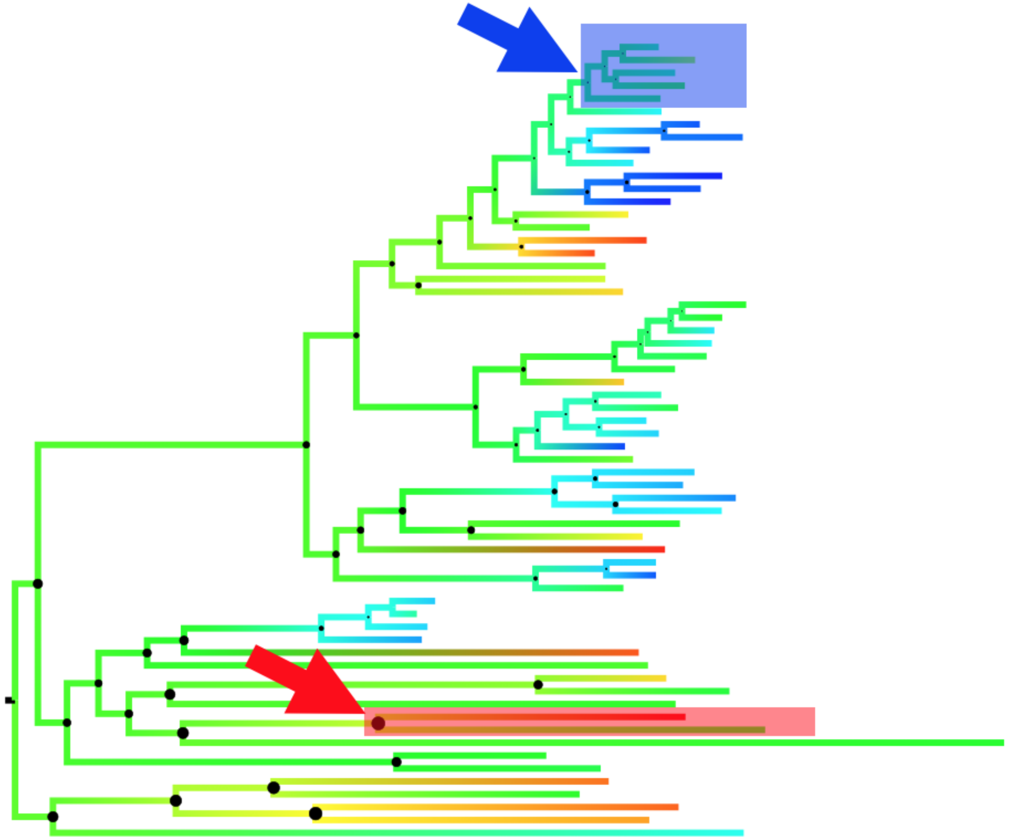
Solving a Hybridization Puzzle


cpDNA capture
Heuchera
Mitella

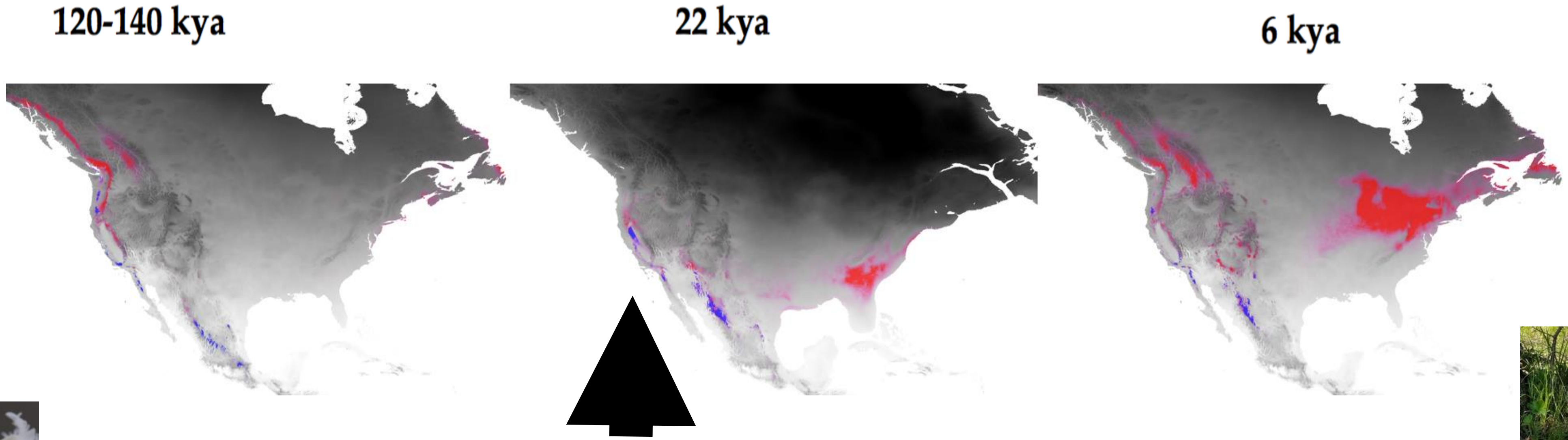
Ryan Folk

Hybridization & Ancestral Ecological Niche-Space

Heuchera *Mitella*



Ancestral Niche Reconstruction


120-140 kya
22 kya
6 kya

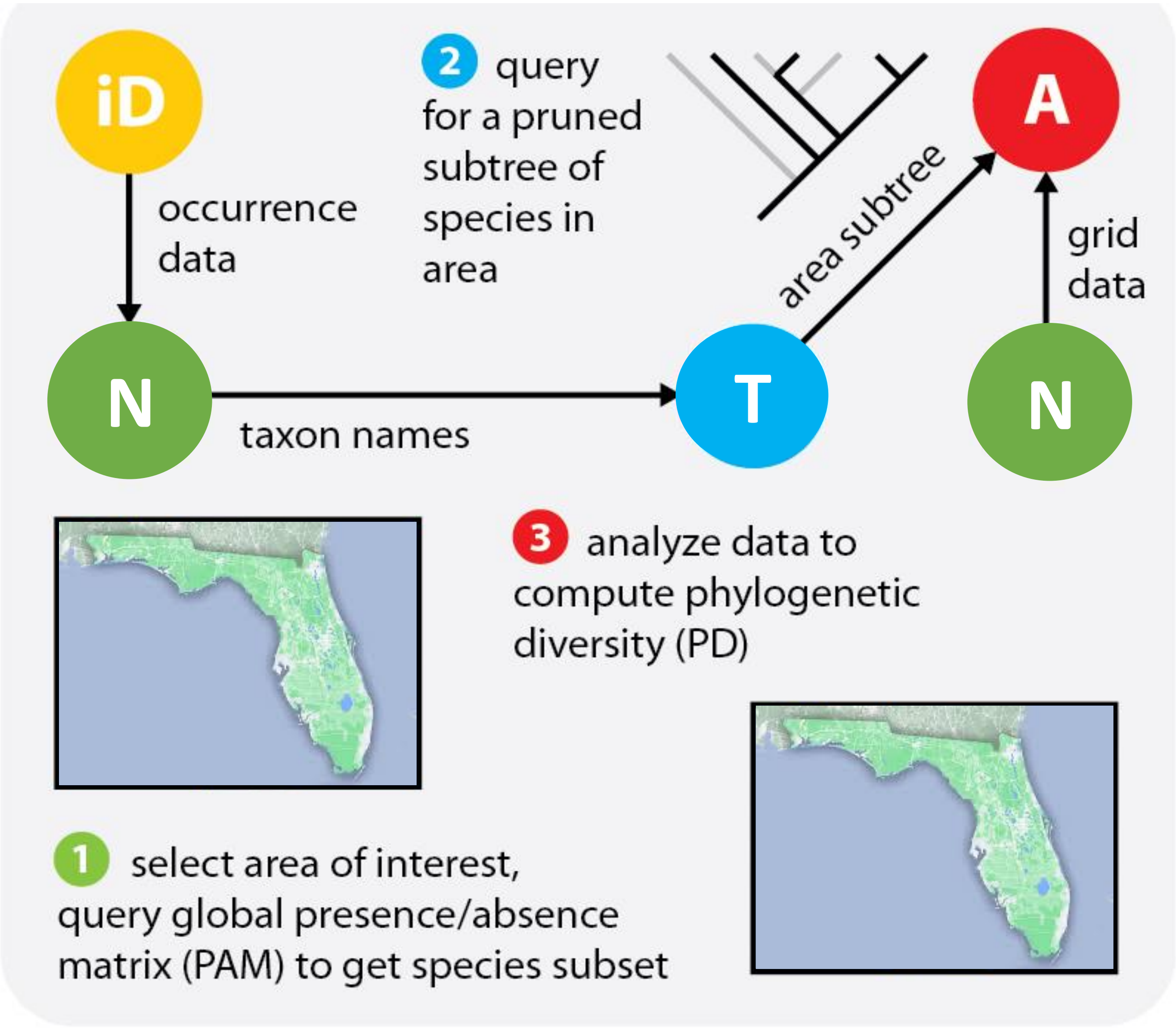
Heuchera
Mitella

Heuchera

Ryan Folk

Folk et al. 2023. Identifying Climatic Drivers of Hybridization with a New Ancestral Niche Reconstruction Method. *Syst Biol.* 2023;72(4):856-873. doi: 10.1093/sysbio/syad018. PMID: 37073863; PMCID: PMC10405357

Spatial Phylogenetics


iD
occurrence data
N
taxon names
T
area subtree
A
grid data
N

2 query for a pruned subtree of species in area

3 analyze data to compute phylogenetic diversity (PD)

1 select area of interest, query global presence/absence matrix (PAM) to get species subset

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iD iDigBio
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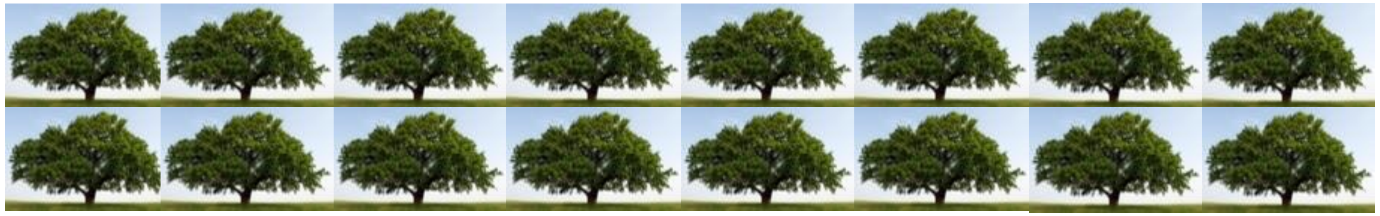
- specimen data / images
- fossil data / images

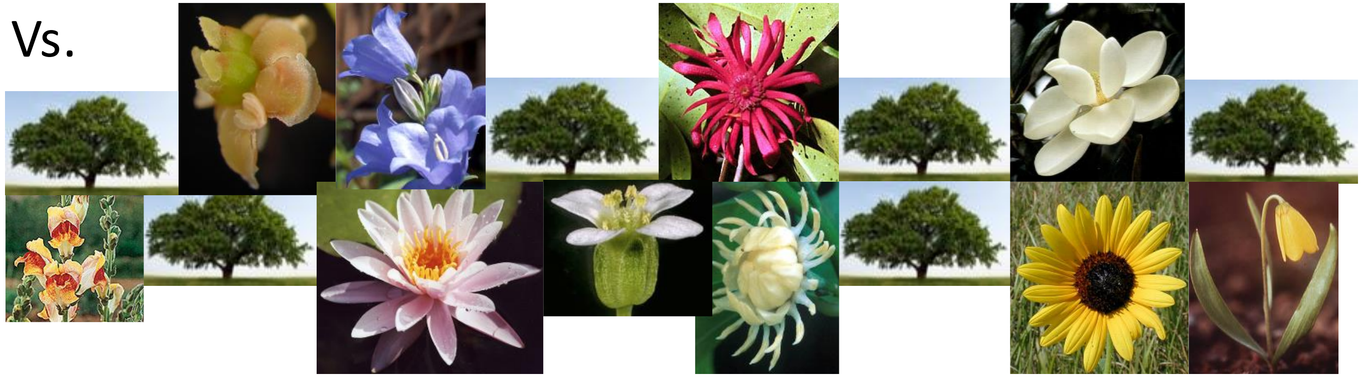
Phylogenetic Diversity (PD):

How much of the Tree of Life is present in an area?

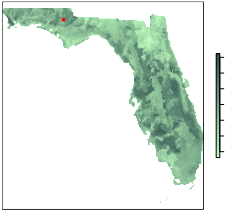
Oaks

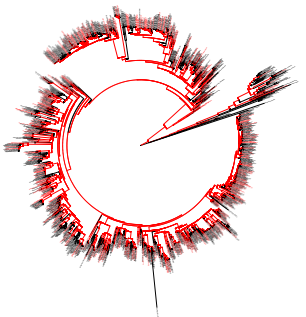


Vs.



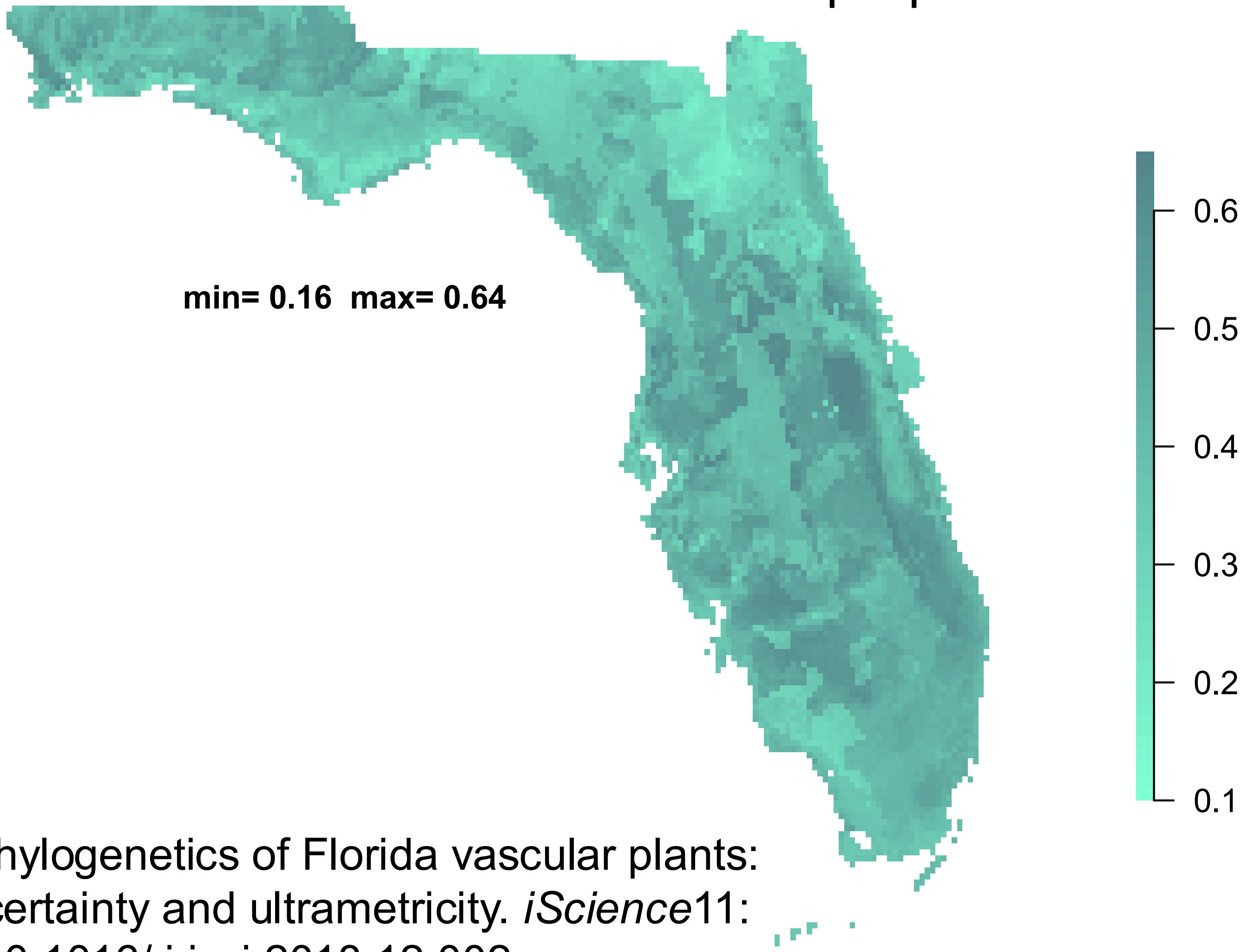
Phylogenetic Diversity (PD):
How much of the Tree of Life is present across Florida?


700
600
500
400
300
200
100



Phylogenetic Diversity

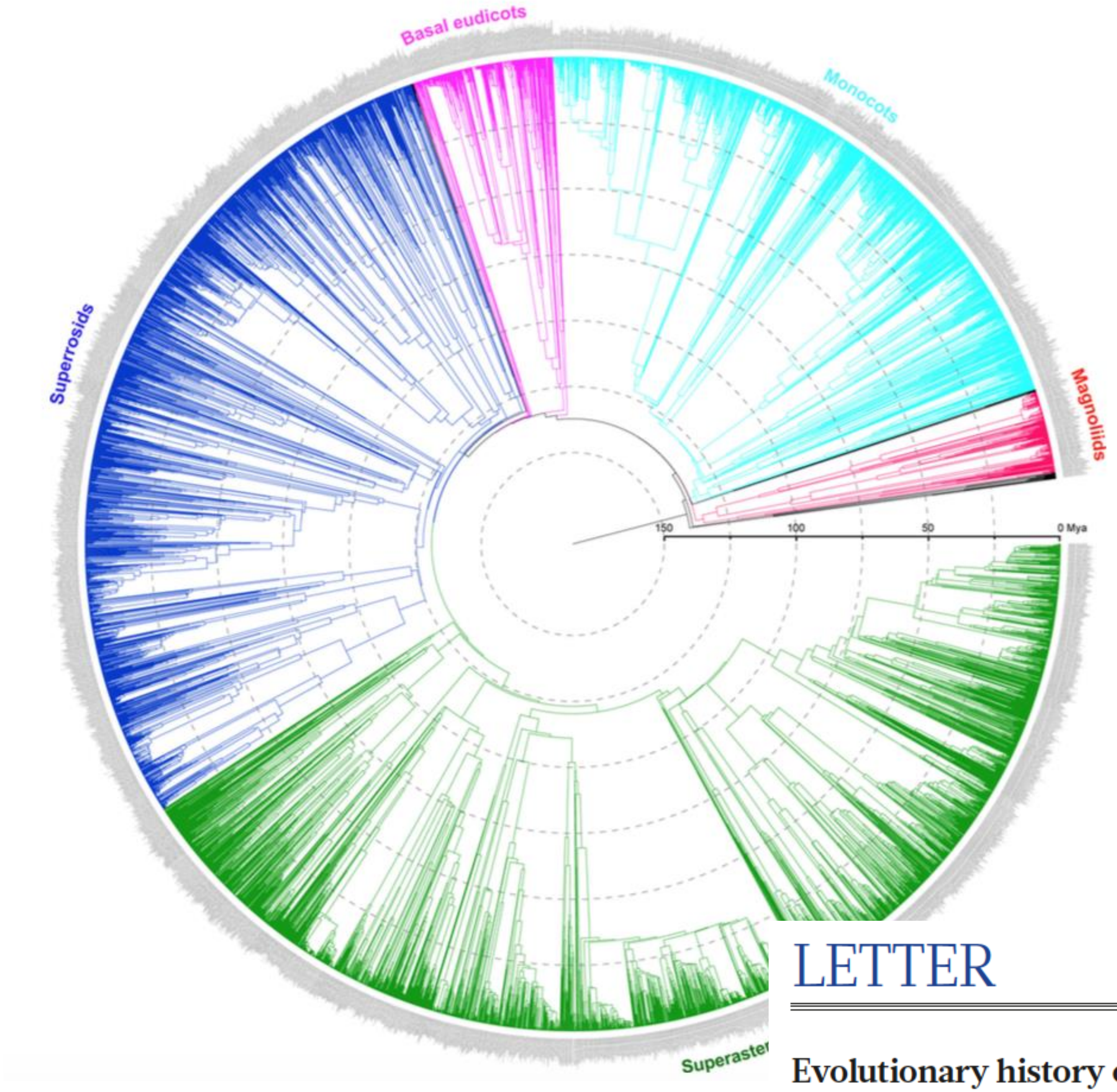
PD = proportion of tree present in cell


min= 0.16 max= 0.64
0.6
0.5
0.4
0.3
0.2
0.1

Allen et al. 2019. Spatial phylogenetics of Florida vascular plants: The effects of tree uncertainty and ultrametricity. *iScience*11: 57–70 https://doi.org/10.1016/ j.isci.2018.12.002

Evolutionary History of the Angiosperm Flora of China

- 30,000 species vascular plants
- 3176 genera
- 92% of genera sampled
- 26,978 species tree


Basal eudicots
Monocots
Magnoliids
Superrosids
Superasterids
150
100
50
0 Mya

Lu et al. *Nature*, 2018

LETTER

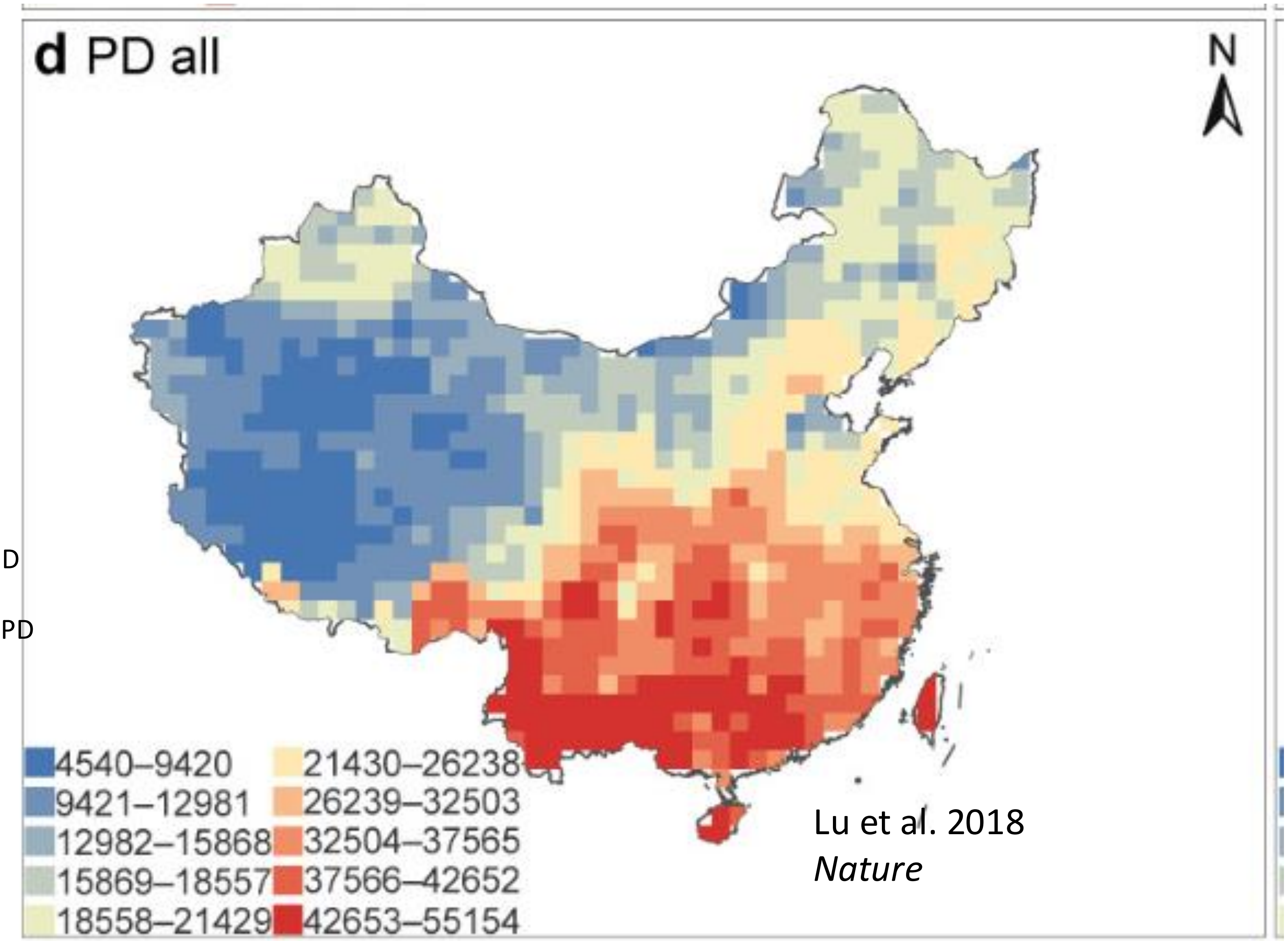
doi:10.1038/nature25485

Evolutionary history of the angiosperm flora of China

Phylogenetic Diversity: China

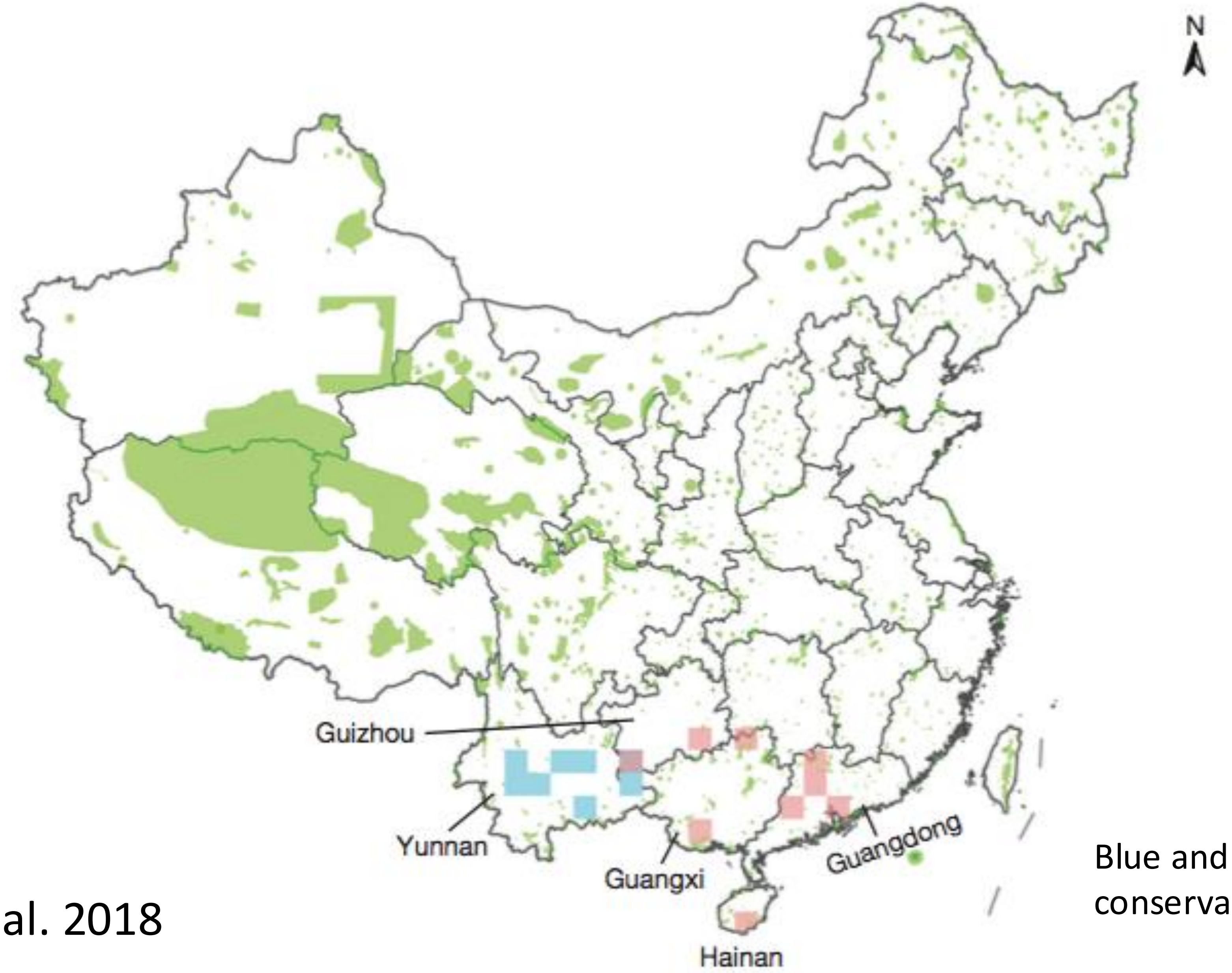
🟦 Low PD

🟥 High PD


d PD all
N
4540–9420
9421–12981
12982–15868
15869–18557
18558–21429
21430–26238
26239–32503
32504–37565
37566–42652
42653–55154

Lu et al. 2018
Nature

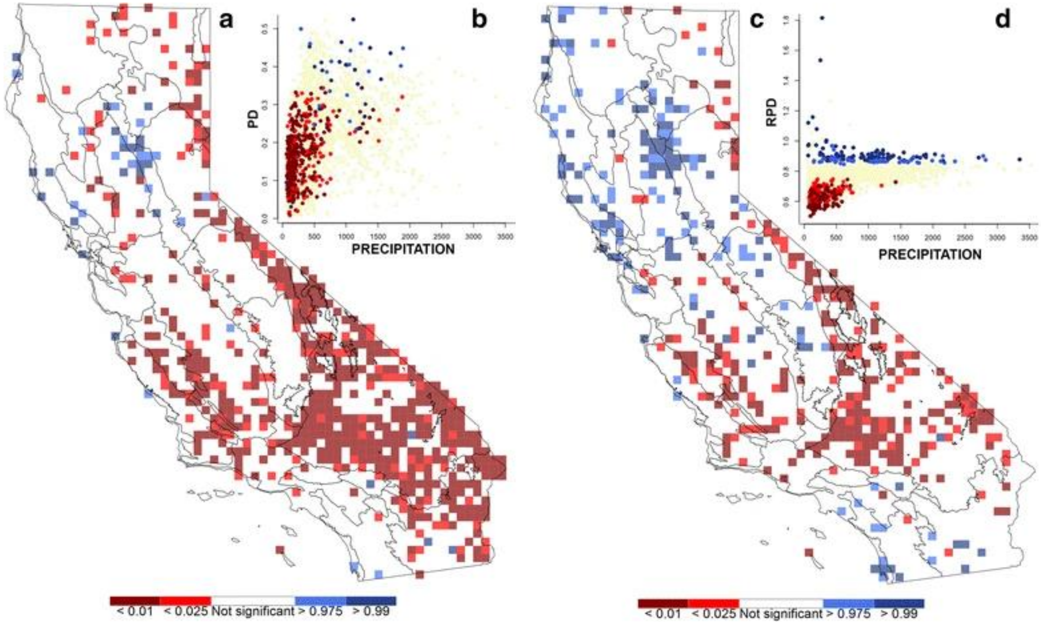
Conservation: Protected Areas vs. High PD in China


N
Guizhou
Yunnan
Guangxi
Guangdong
Hainan

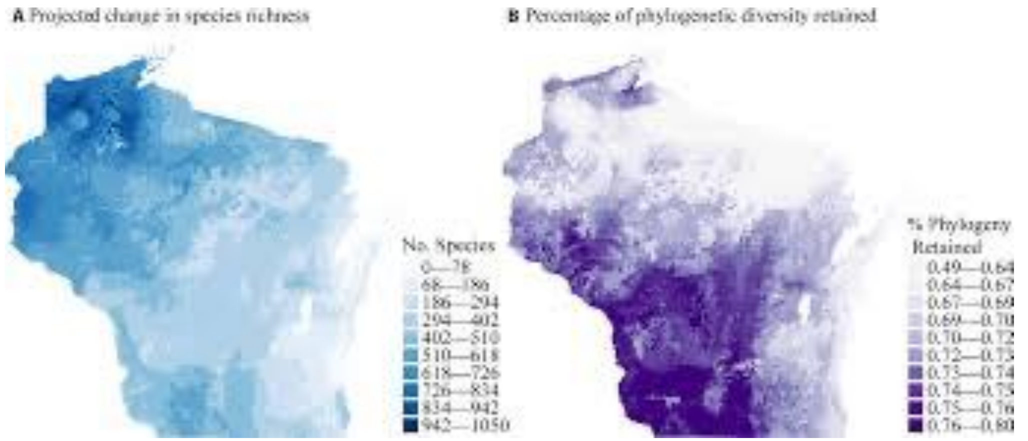
Blue and Pink =
conservation areas

Lu et al. 2018

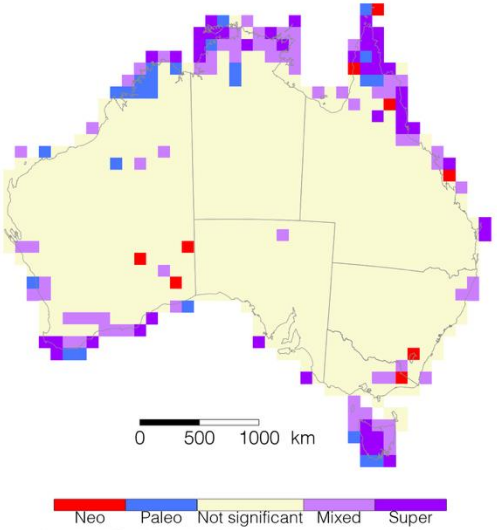
Phylogenetic Diversity


a
b
PD
PRECIPITATION
c
d
RPD
PRECIPITATION
< 0.01 < 0.025 Not significant > 0.975 > 0.99
< 0.01 < 0.025 Not significant > 0.975 > 0.99

Thornhill *et al.* Spatial phylogenetics of the native California flora. *BMC Biol* **15**, 96 (2017). https://doi.org/10.1186/s12915-017-0435-x


A Projected change in species richness
B Percentage of phylogenetic diversity retained
No. Species
0—78
68—186
186—294
294—402
402—510
510—618
618—726
726—834
834—942
942—1050
% Phylogeny Retained
0.49—0.64
0.64—0.67
0.67—0.69
0.69—0.70
0.70—0.72
0.72—0.73
0.73—0.74
0.74—0.75
0.75—0.76
0.76—0.80

Spalink, D. et al. 2018. Spatial phylogenetics reveals evolutionary constraints on the assembly of a large regional flora. *American Journal of Botany* 105(11): 1938–19

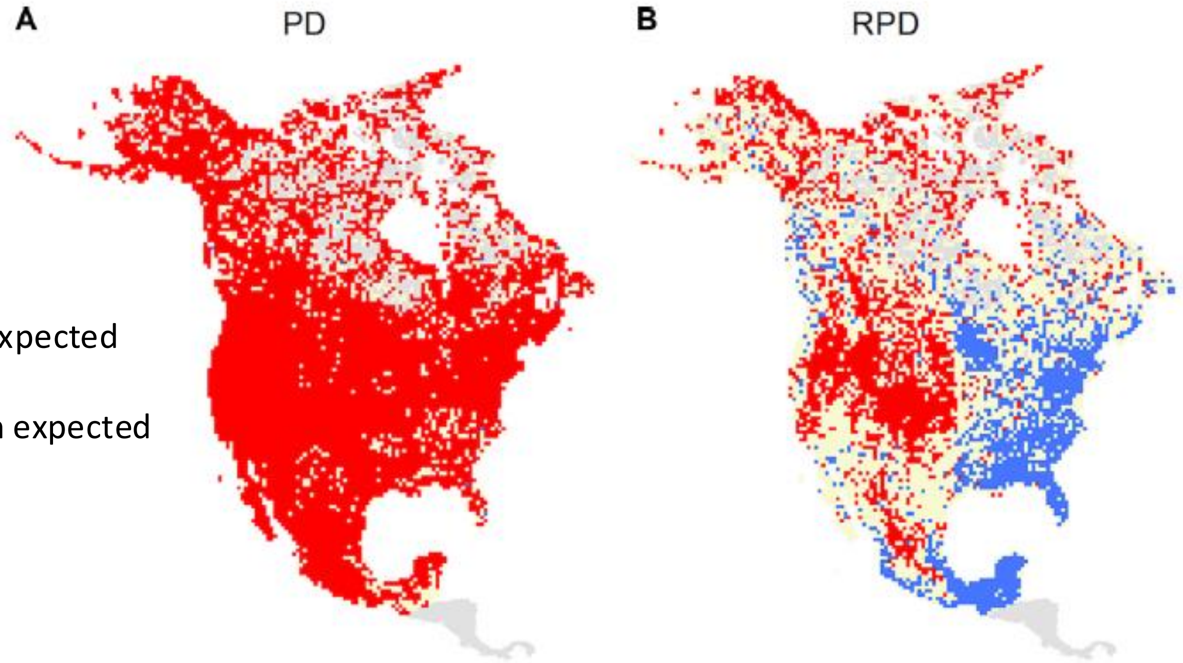

0
500
1000 km
Neo
Paleo
Not significant
Mixed
Super

Mishler *et al.* Phylogenetic measures of biodiversity and neo- and paleo-endemism in Australian *Acacia*. *Nat Commun* **5**, 4473 (2014). https://doi.org/10.1038/ncomms5473

Spatial Phylogenetics of the North American Flora

Red = significantly less PD than expected

Blue = significantly more PD than expected



Red = significant concentration of short branches

Blue = significant concentration of long branches

Fig. 4. A, PD significance with an unconstrained randomization. Virtually the entire continent is significantly low in PD, meaning that taxa are more closely related than expected by chance. **B,** RPD significance with an unconstrained randomization. Areas in blue have a concentration of significantly longer branches than expected; areas in red have a concentration of significantly shorter branches than expected. There is a striking difference between the western and eastern parts of the continent. PD, phylogenetic diversity; RPD, relative phylogenetic diversity.

Mishler et al. 2020. J. Syst. Evol., 58: 393-405. https://doi.org/10.1111/jse.12590

Relative Phylogenetic Diversity

RPD - *relative phylogenetic diversity=*

$$\frac{\text{PD on the original tree}}{\text{PD on a comparison tree with all branch lengths equal}}$$

significant concentration of short branches

VS

significant concentration of long branches

RPD: identify and distinguish areas of phylogenetic overdispersion and clumping that reflect signals of biogeographic history and ecological processes

Trait Evolution

RESOURCES:

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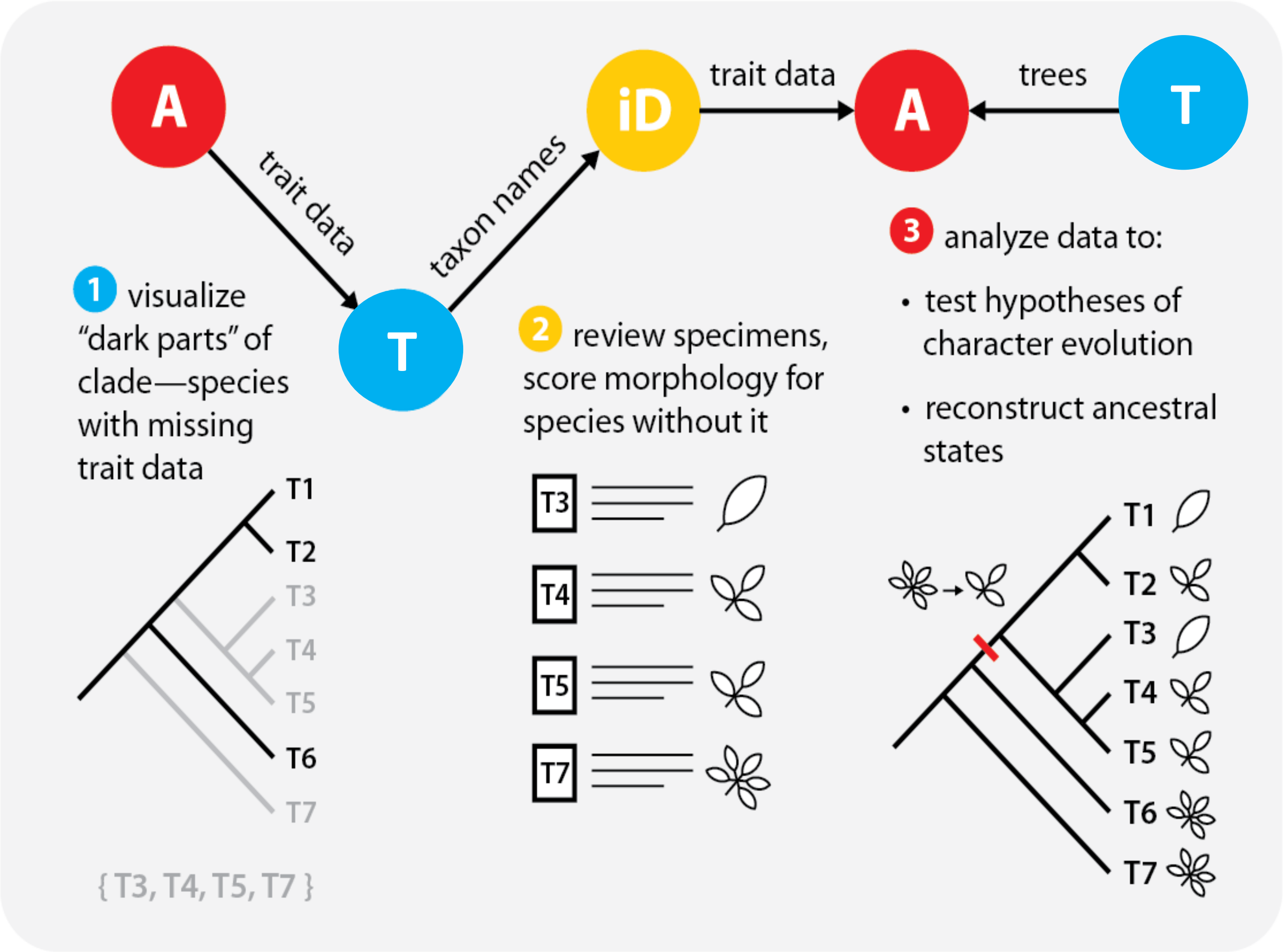
- fossil data / images


A
trait data
T
taxon names
iD
trait data
A
trees
T
T1
T2
T3
T4
T5
T6
T7
{ T3, T4, T5, T7 }
T3
T4
T5
T7
T1
T2
T3
T4
T5
T6
T7

1 visualize "dark parts" of clade—species with missing trait data

2 review specimens, score morphology for species without it

3 analyze data to:
- test hypotheses of character evolution
- reconstruct ancestral states

Ancestral Niche

Connecting Trees, Specimens, Tools

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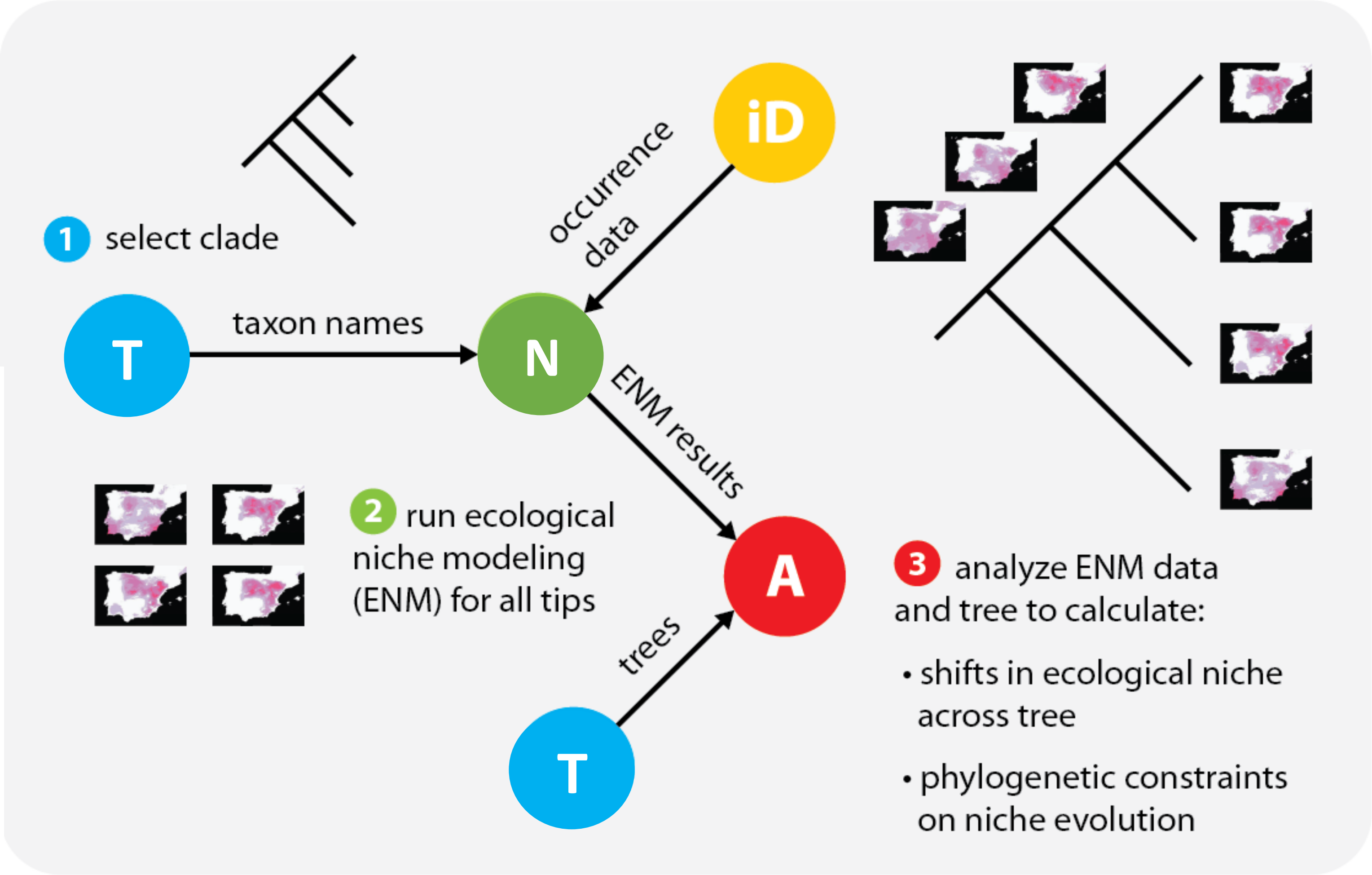
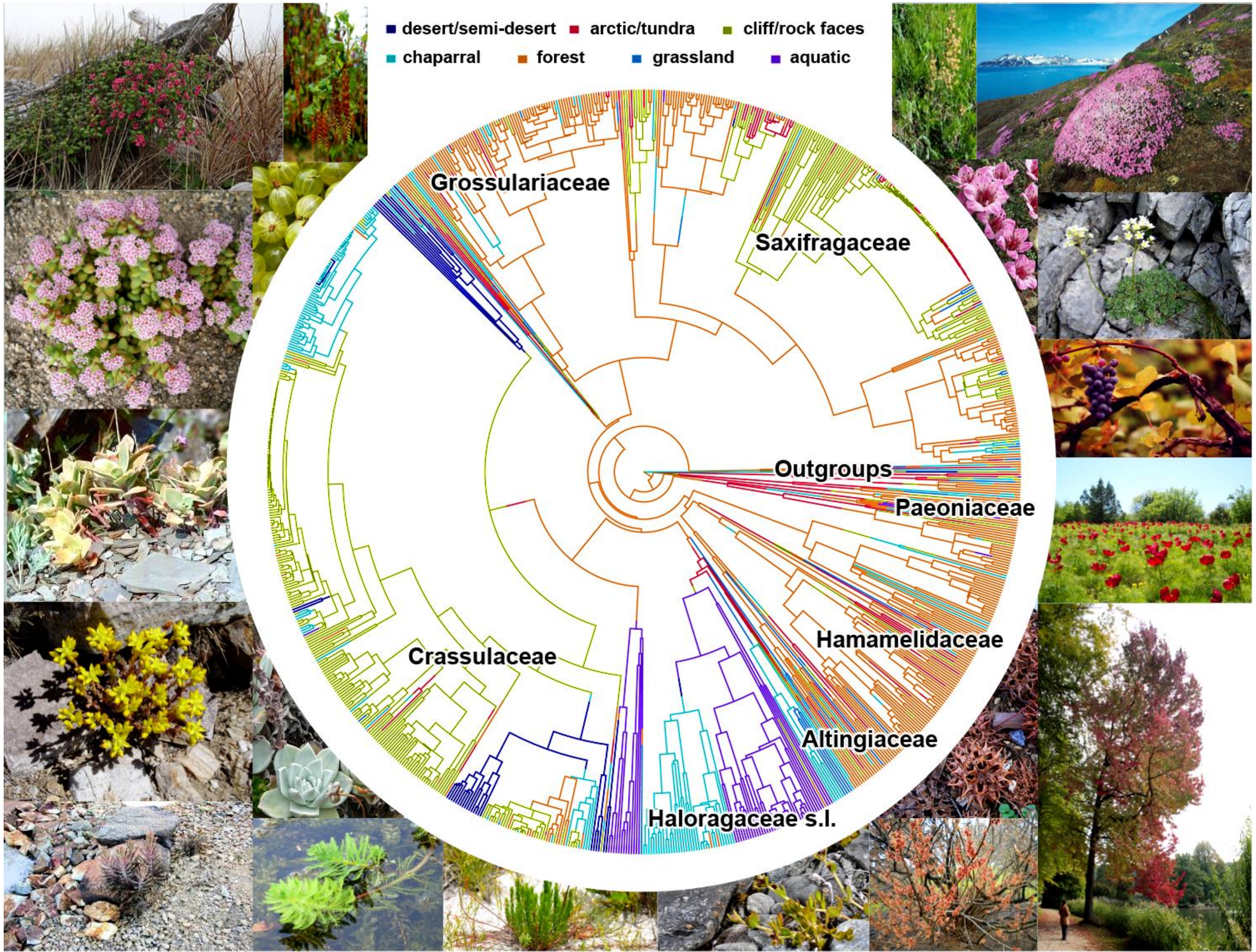


1 select clade

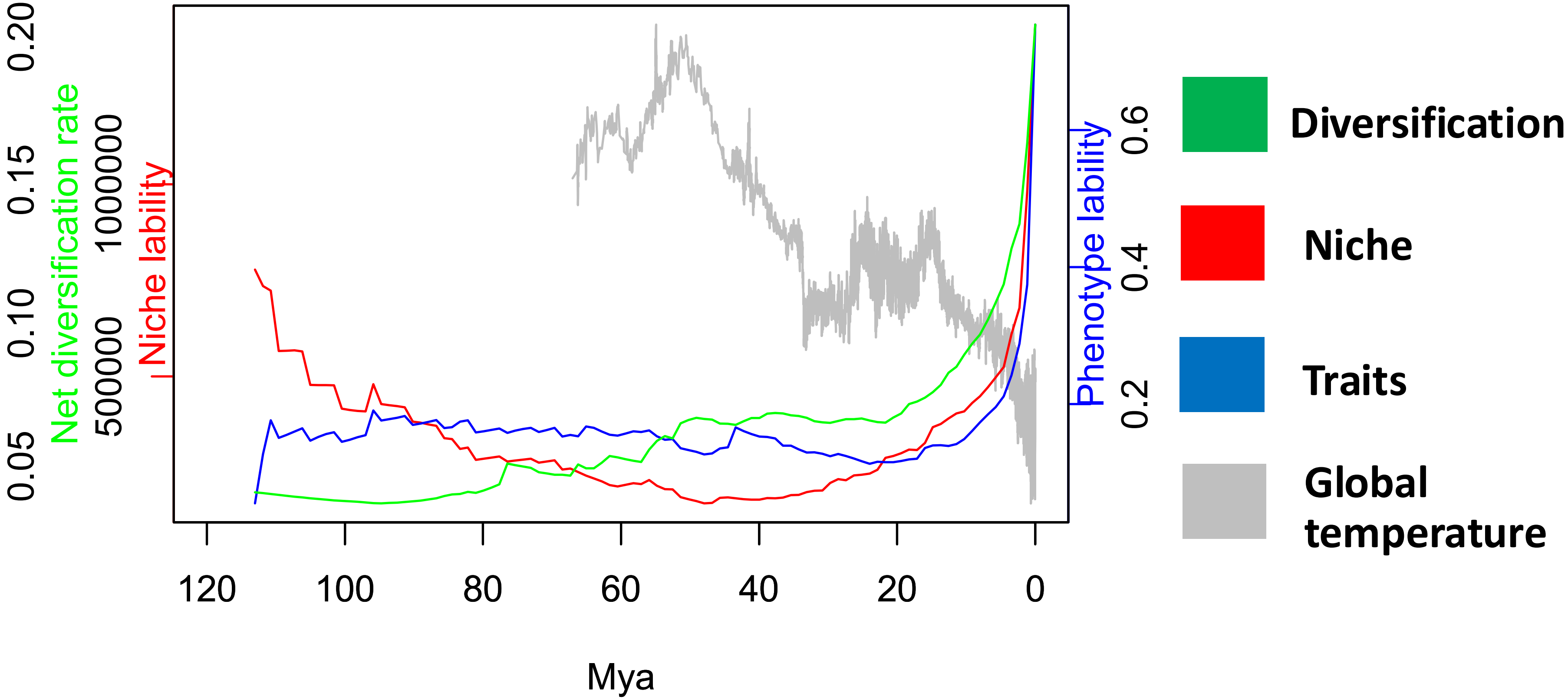
2 run ecological niche modeling (ENM) for all tips

3 analyze ENM data and tree to calculate:
- shifts in ecological niche across tree
- phylogenetic constraints on niche evolution

Saxifragales: Hyper-diverse


■ desert/semi-desert
■ arctic/tundra
■ cliff/rock faces
■ chaparral
■ forest
■ grassland
■ aquatic
Grossulariaceae
Saxifragaceae
Outgroups
Paeoniaceae
Hamamelidaceae
Altingiaceae
Haloragaceae s.l.
Crassulaceae

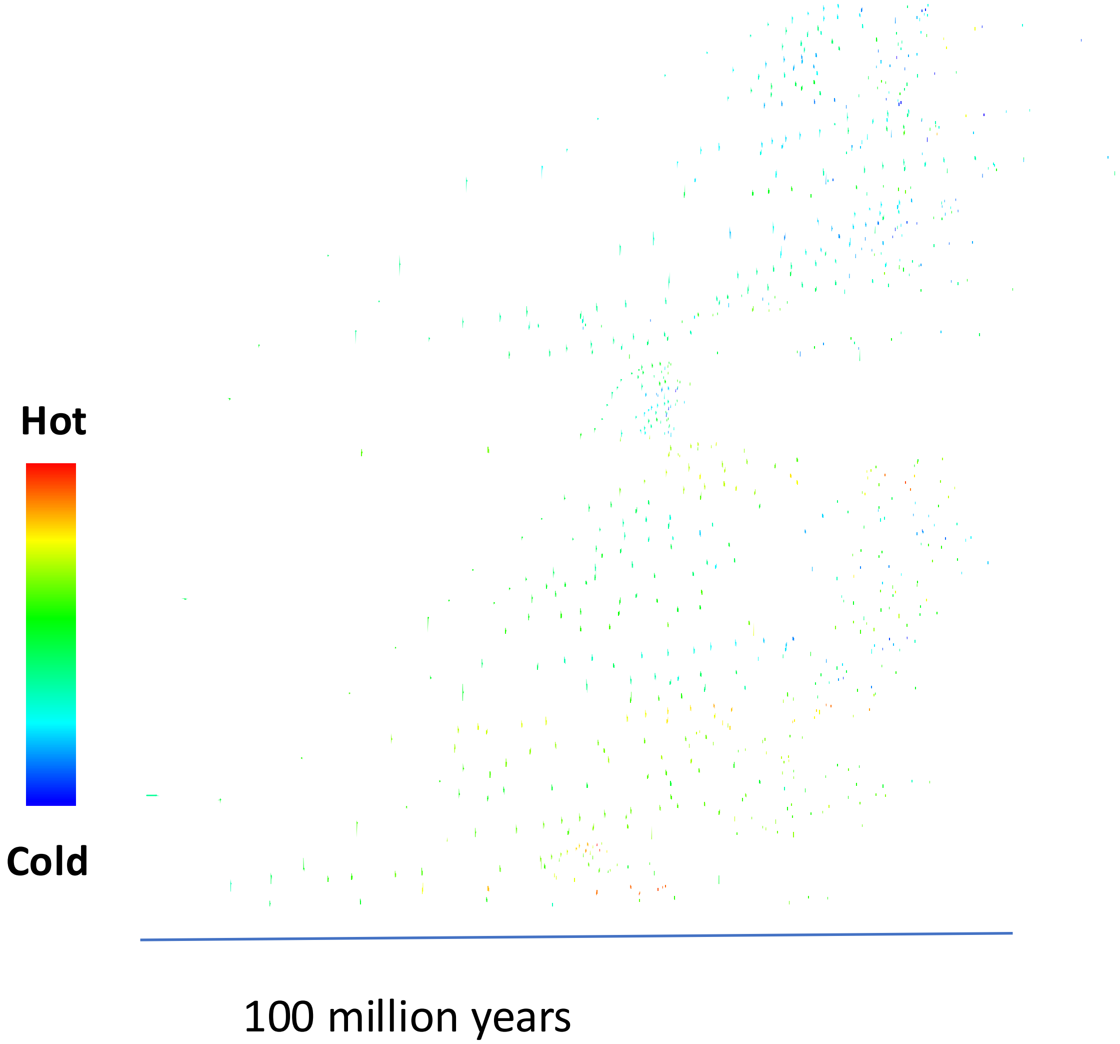
A Temperate Radiation: Mid-Miocene Cooling



Folk et al. 2019. Uncoupled diversification, ecological niche and phenotype in a temperate radiation. *PNAS* 116: 10874-10882.

Response of Saxifragales to Climate Change

- **Mean annual temperature**
- Shifts are rare


Hot
Cold
100 million years

Ryan Folk

Relative Phylogenetic Diversity and Relative Phylogenetic Endemism

RPD is ratio of A/C:

RPD - *relative phylogenetic diversity=*

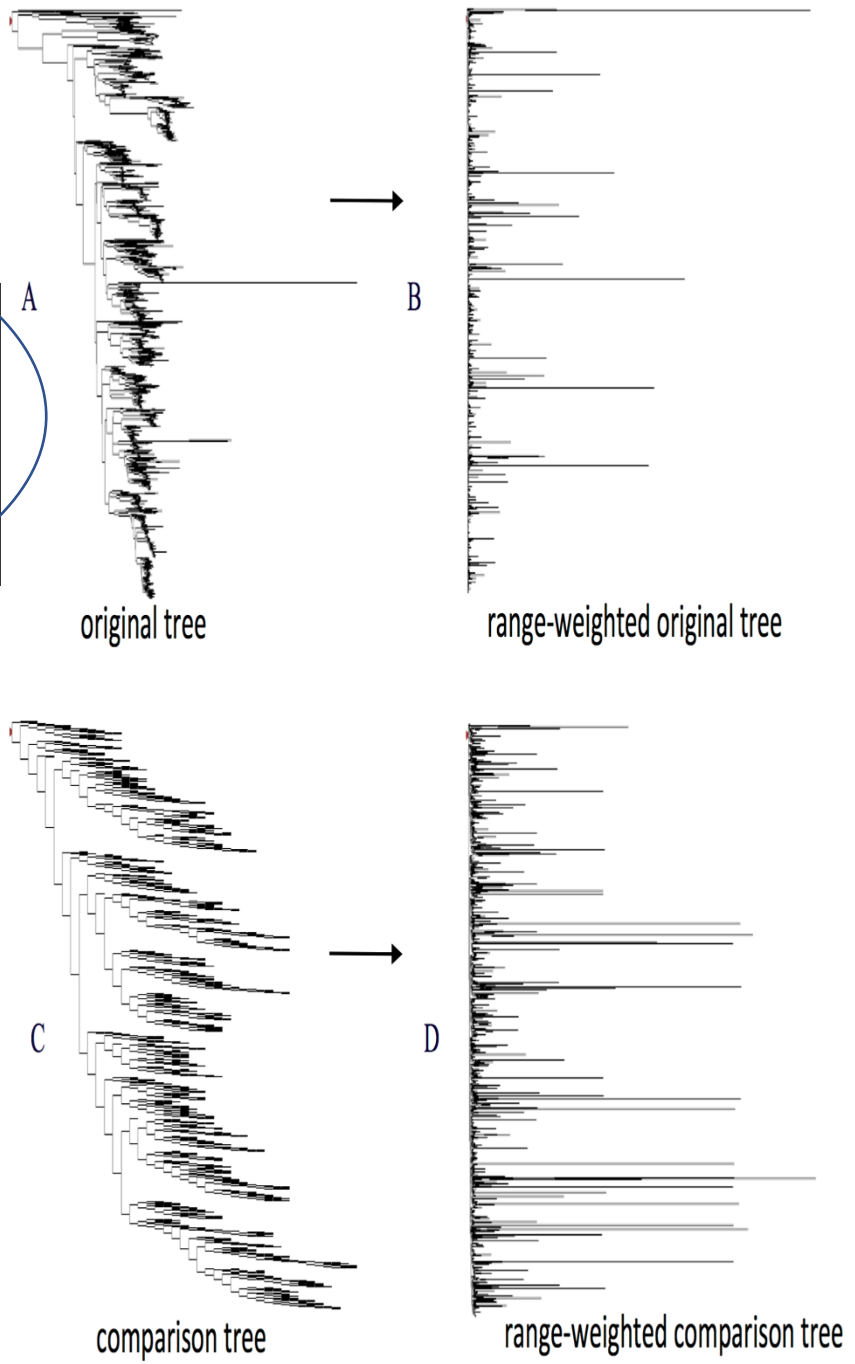
$$\frac{\text{PD on the original tree}}{\text{PD on a comparison tree with all branch lengths equal}}$$

RPE is ratio of B/D:

RPE - *relative phylogenetic endemism =*

$$\frac{\text{PD on the range-weighted original tree}}{\text{PD on a range-weighted comparison tree with all branch lengths equal}}$$

Mishler et al., (2014)


A
B
original tree
range-weighted original tree
C
D
comparison tree
range-weighted comparison tree

Thanks